

ENTITY RELATIONSHIP (ER) DIAGRAM AND RELATIONAL SCHEMA

AIRLINE RESERVATION SYSTEM

1. Objective

The selected application is an **Airline Reservation System**. The database stores passenger information, booking details, flight information, aircraft details, and payment information. The system allows passengers to make bookings for flights and maintains information about the aircraft operating each flight. It also records the payment associated with each booking. The design represents the **many-to-many relationship between passengers and flights through the BOOKING entity**.

2. Entities and Attributes

The following entities are used in the ER model. **PK** denotes Primary Key and **FK** denotes Foreign Key.

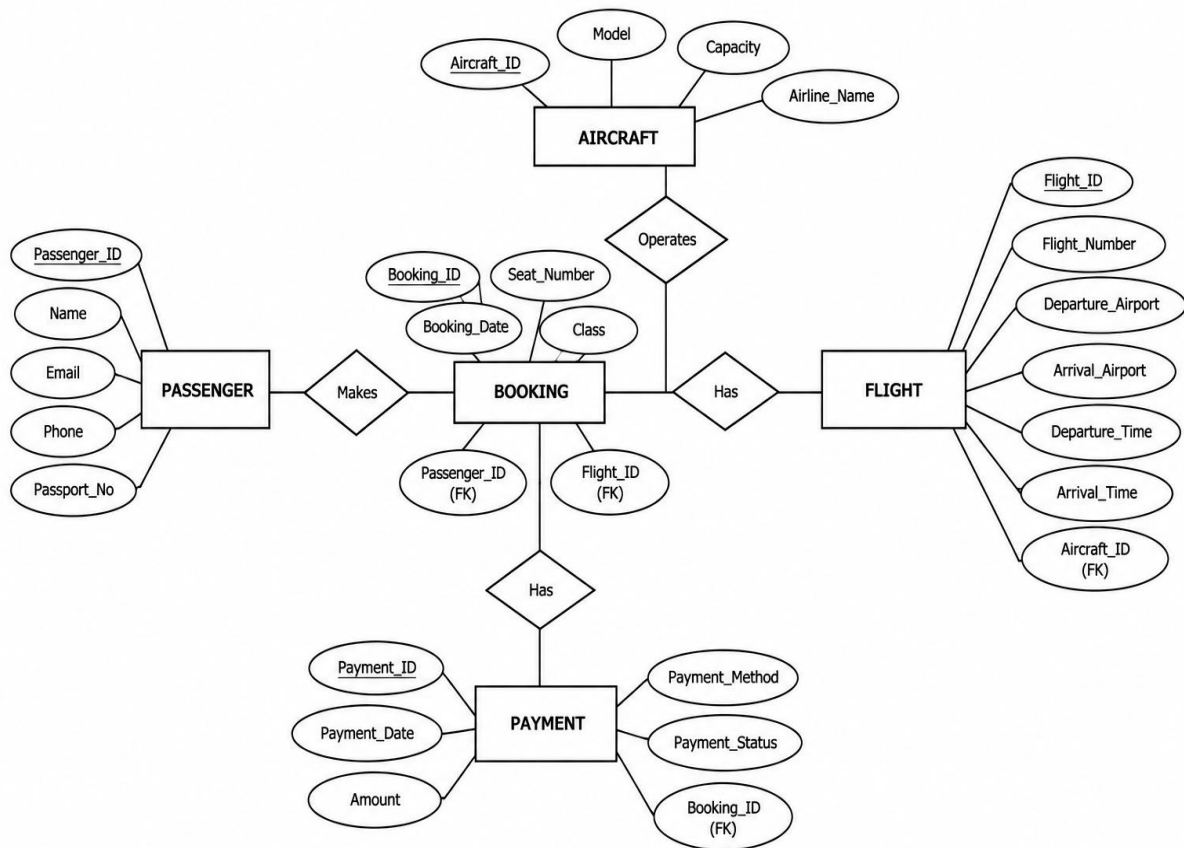
Entity	Primary Key	Other Attributes / Foreign Keys
PASSENGER	passenger_id (PK)	name, email, phone, passport_no
BOOKING	booking_id (PK)	booking_date, seat_number, class, passenger_id (FK), flight_id (FK)
FLIGHT	flight_id (PK)	flight_number, departure_airport, arrival_airport, departure_time, arrival_time, aircraft_id (FK)
AIRCRAFT	aircraft_id (PK)	model, capacity, airline_name
PAYMENT	payment_id (PK)	payment_date, amount, payment_method, payment_status, booking_id (FK)

Entity Description

- **PASSENGER** – Stores information about passengers.
- **BOOKING** – Stores reservation details made by passengers for flights.
- **FLIGHT** – Stores information about scheduled flights.
- **AIRCRAFT** – Stores information about aircraft used for flights.
- **PAYMENT** – Stores payment details associated with bookings.

3. ER Diagram

The ER diagram below represents the entities, attributes, relationships, and overall structure of the **Airline Reservation System**.



4. Relationships and Cardinalities

Entities	Relationship	Cardinality	Explanation
PASSENGER — BOOKING	Makes	1 : M	One passenger can make many bookings, while each booking belongs to one passenger.
FLIGHT — BOOKING	Has	1 : M	One flight can have many bookings, while each booking is associated with one flight.
AIRCRAFT — FLIGHT	Operates	1 : M	One aircraft can operate multiple flights, while each flight is operated by one aircraft.
BOOKING — PAYMENT	Has	1 : 1	Each booking has one payment record, and each payment is associated with one booking.
PASSENGER — FLIGHT	Books	M : N	A passenger can book many flights and a flight can have many passengers. The BOOKING entity resolves this relationship.

5. Relational Schema

The ER model is converted into relations using standard mapping rules. Entities become tables, primary keys uniquely identify records, and foreign keys represent relationships between tables. The M:N relationship between passengers and flights is resolved through the BOOKING table.

Relation	Relational Schema
PASSENGER	PASSENGER(passenger_id PK, name, email, phone, passport_no)
AIRCRAFT	AIRCRAFT(aircraft_id PK, model, capacity, airline_name)
FLIGHT	FLIGHT(flight_id PK, flight_number, departure_airport, arrival_airport, departure_time, arrival_time, aircraft_id FK)
BOOKING	BOOKING(booking_id PK, booking_date, seat_number, class, passenger_id FK, flight_id FK)
PAYMENT	PAYMENT(payment_id PK, payment_date, amount, payment_method, payment_status, booking_id FK)

6. Primary Key and Foreign Key Identification

Table	Primary Key (PK)	Foreign Key(s) (FK)
PASSENGER	passenger_id	—
AIRCRAFT	aircraft_id	—
FLIGHT	flight_id	aircraft_id → AIRCRAFT(aircraft_id)
BOOKING	booking_id	passenger_id → PASSENGER(passenger_id); flight_id → FLIGHT(flight_id)
PAYMENT	payment_id	booking_id → BOOKING(booking_id)

7. Brief Relationship Explanation

- Passenger–Booking: The relationship is 1:M because one passenger may make several bookings, while each booking is associated with one passenger.
- Flight–Booking: The relationship is 1:M because one flight may have multiple bookings, while each booking refers to one flight.

- Aircraft–Flight: The relationship is 1:M because an aircraft can operate multiple flights, while each flight is assigned to one aircraft.
- Booking–Payment: The relationship is 1:1 because each booking has one payment record, and each payment belongs to one booking.
- Passenger–Flight: The relationship is M:N because one passenger can book multiple flights and one flight can have multiple passengers. The BOOKING entity acts as the associative entity that resolves this M:N relationship.

8. ERD Reference

